



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KPK THILLAI AGENCI ESS IOC DEALERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

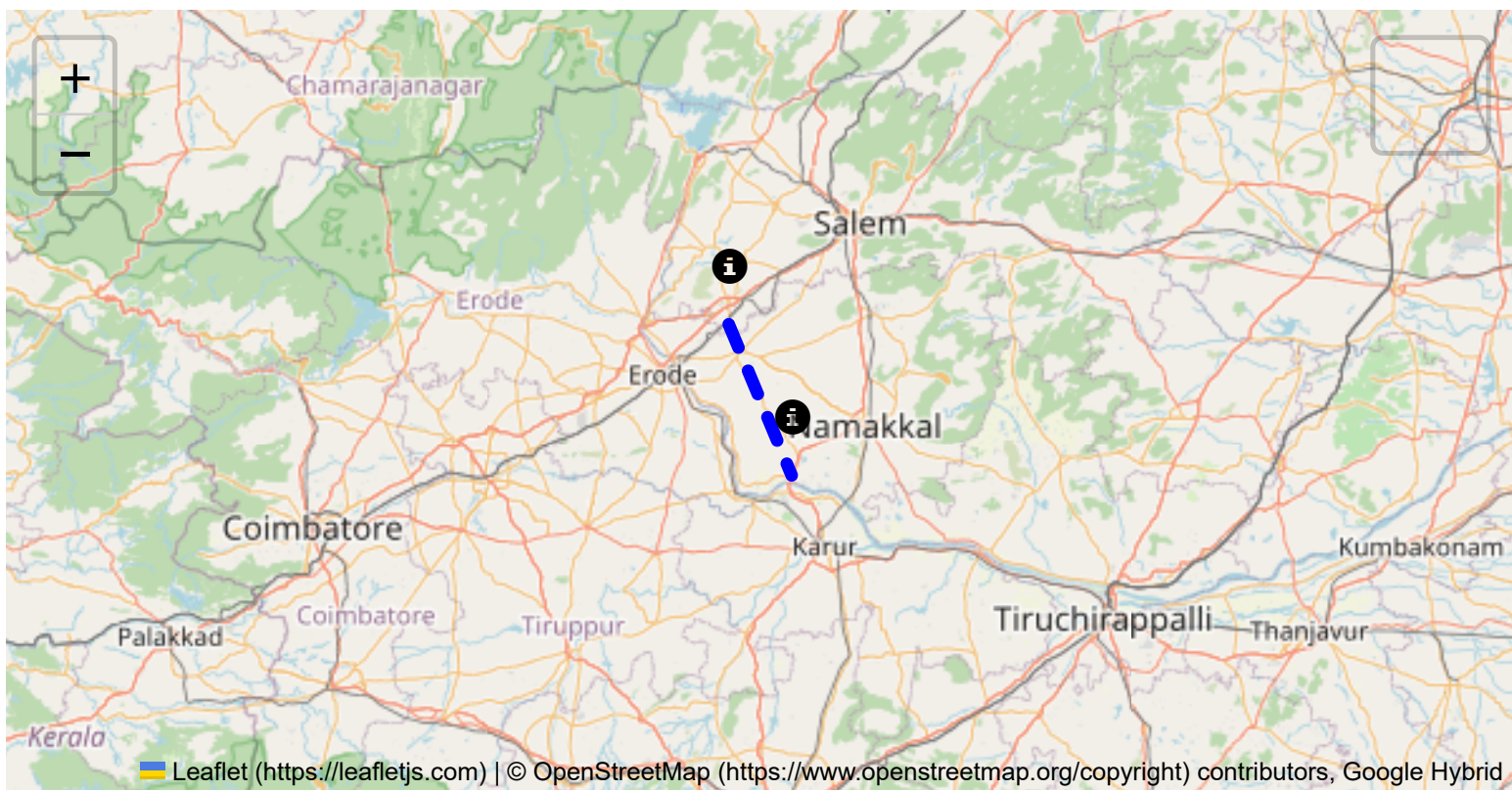
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

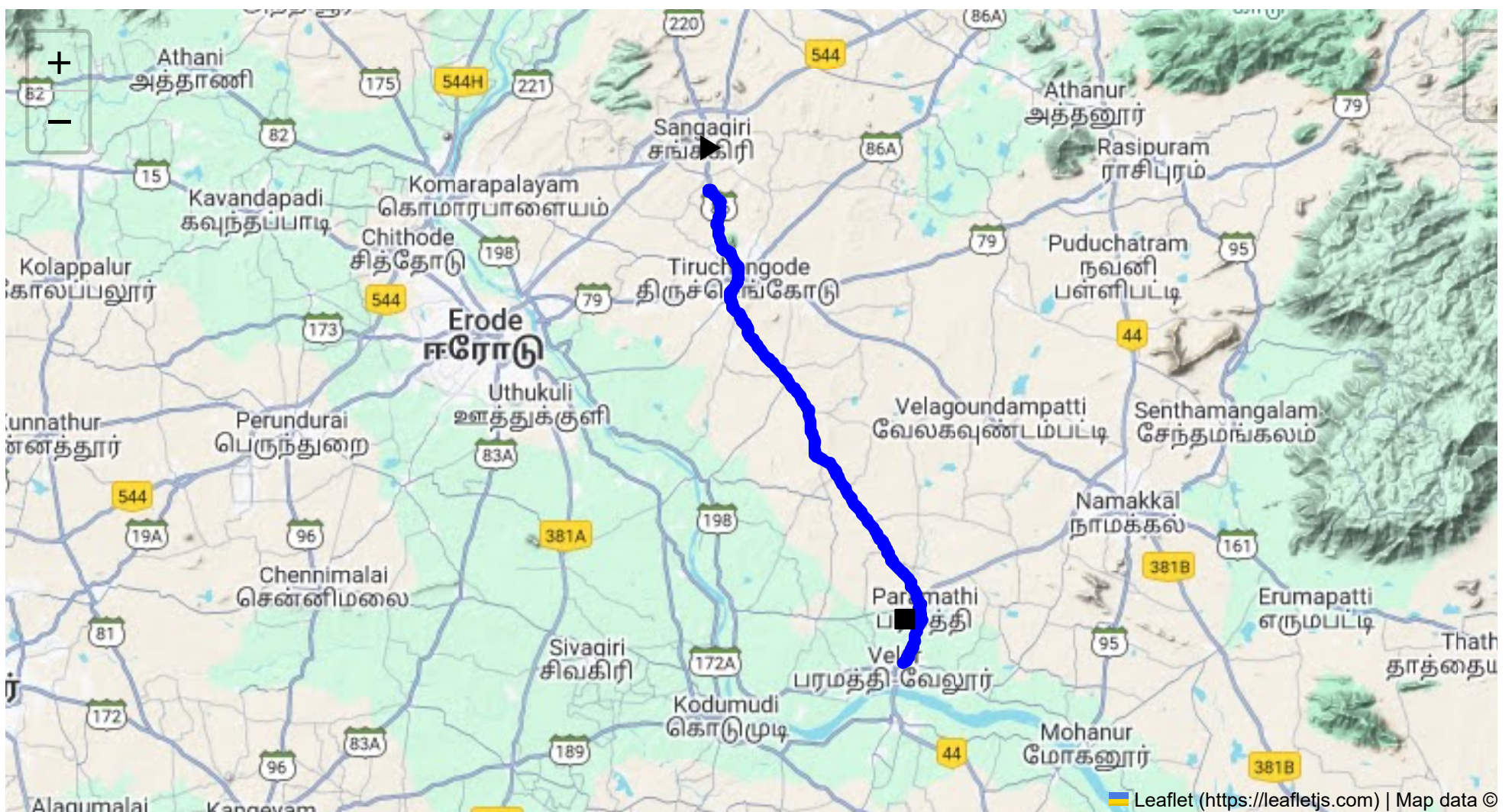
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.88 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.115649, 78.009621)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 4285+7R3, Anna Salai, Velur covers approximately 42.88 kilometers. It involves traveling mostly on state highways and some rural roads. The main road connections would typically include local highways and possibly sections of NH544, if the approach

requires such adjustments. The route passes through semi-urban to rural settings, dotted with small towns and agricultural fields.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with three distinct seasons: summer, monsoon, and winter. The area may face heavy rains during the Northeast Monsoon from October to December, leading to potential flooding and reduced visibility. High temperatures can be expected from March to May, often surpassing 35°C, posing risks of heat-related vehicle issues.

3. Analysis of Traffic Patterns

Traffic congestion in this region mainly peaks during the morning (7-9 AM) and evening (5-7 PM) due to local commuter traffic. The roads around small towns and markets can become particularly congested. Weekend traffic might show variability due to local events or market days.

4. Assessment of Road Quality and Infrastructure

The state highways in this region are generally well-maintained, but rural roads may have uneven surfaces, potholes, or narrow lanes, requiring cautious driving. Signage might be sparse in remote stretches, and lighting is limited, which can affect nighttime visibility.

5. Suggestions for Alternative Routes

In emergencies, routes bypassing congested areas or potential natural hazard zones should be considered. Utilizing local advisories or navigation apps that provide real-time updates on road conditions and closures will be essential.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires compliance with local traffic rules and road safety standards as outlined by the Motor Vehicles Act. Specific permits might be required, especially when transporting through populated areas. Drivers should be aware of restrictions on movement during peak hours or times of public events.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been instances of accidents involving heavy vehicles on the main highways due to speeding or weather conditions. However, detailed records specific to this route are scarce. Investigating further through local transport authorities would provide insights into specific trends or incidents.

8. Environmental Considerations and Sensitive Areas

The route intersects agricultural lands and potentially crosses small water bodies, necessitating adherence to environmental regulations to minimize spill risks. Awareness of wildlife or livestock crossings is important in rural stretches.

9. Analysis of Communication Coverage

Most of Tamil Nadu has good mobile network coverage; however, there could be intermittent dead zones in more remote areas, especially in rural sections. Drivers should prepare for potential communication gaps.

10. Estimated Emergency Response Times

Emergency response in urban areas or near towns might range from 15 to 30 minutes, though rural sections could experience delays, extending response times up to 45 minutes or more due to lack of immediate resources and personnel.

12. Overall Summary of Risk Assessment

The route is generally safe with manageable risks related to typical weather fluctuations and limited infrastructure challenges. The primary concerns involve potential weather disruptions, traffic congestion during peak times, and variations in road quality. Adhering to local regulations, preparing for possible communication outages, and practicing safe driving in rural areas will mitigate many risks. Continuous monitoring of road conditions and maintaining updated emergency contacts are advisable for the safe transit of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	11.12555, 78.01664	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	High	11.12540, 78.01613	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12555, 78.01664



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12540, 78.01613
